



**Faculty of Information and Communication Technology
Mahidol University**

**Project 2: Conceptual Database Design
E-Commerce: IKEA**

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ITCS442 Relational Database Design
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Business Domain and Problem statement

Real Business Company: IKEA (Customer and Order Management System)

IKEA Thailand provides an omnichannel retail service including online ordering, click & collect, delivery service, installation service, assembly service, etc. Customers can use the IKEA website to track their orders and schedule to access the services, while IKEA staff are responsible for the process such as preparing orders, scheduling deliveries, and arranging booking services. However, operational data for each section is separated across different systems and workflows, such as tracking delivery, furniture assembly services, and other services. Each service has different conditions depending on the specifications of the product, resulting in an increased amount of data and the need for continuous status updates.

Problem statement

This project proposes the development of a centralized database application that combines order fulfillment, service booking, and return management into a single system. This system aims to reduce operational errors of scattered data and improve data connectivity across different services.

Declaration of AI tool usage : AI Level 2

This project used AI tools (ChatGPT, Gemini) as a supporting assistant for ideas, improving English writing (paraphrasing, grammar correction, and clarity), and organizing report structure.

All database design decisions (e.g. key characteristics, mission statement, mission object, scope and boundary diagram, business function, and requirement specification) were reviewed and verified by everyone in group 1. The final part of analysis, implementation, and report in phase 1 were completed and finalized by team members. We take full responsibility for the accuracy, originality, and integrity of the submitted work

Data Requirements

1. Ordering management

Product

Data	Description
Product_SerialNumber	An unique identifier for each product.
Product_Name	A name of the product for show to customers.
Product_Description	Details of the product used for specification explaining
Product_Measurement	Product size/dimensions used for storage and delivery planning.

Customer

Data	Description
Customer_account_ID	Unique identification for each Customer account.
Customer_Name (1. Customer_FirstName, 2. Customer_LastName)	Customer identity information for customer care and service support.
Address	Address information for delivery, service and account records.
Phone Number	Contact information for customer support communication.
Email	Contact information for updated information.

Order

Data	Description
Order Id	A unique transaction number generated at the time of checkout.
Date	The timestamp when the order was officially placed.
Status	The current stage of the order.
Order_quantity	Total count of unique product lines included in the entire transaction.
logs_history	A chronological record of internal changes.
Total price	The final sum of all products, taxes, and service fees minus any discounts.

Orderline

Data	Description
Product_line_number	A sequence number within a single order.
Amount	The unit price of the product at the specific time of purchase.
Date	Records when that specific line item was added or modified.

Price	The price for each customer order.
Total_price	The calculated subtotal for that specific line

ServiceBooking (Weak Entity of Order)

Data	Description
Booking Id	Unique identifier for the service appointment.
Booking Date	Scheduled day the service is intended to take place.
Booking Status	Current state of the service.
Service Address	The location where the assembly or service will be performed.
TimeSlot	The specific window of time reserved for the service.

Staff

Data	Description
Staff Id	Unique employee identification number.
Staff Name	Full name of the employee.
Phone Number	Work contact number for internal coordination.
Email	Corporate email address for communication and system login.

Shipment

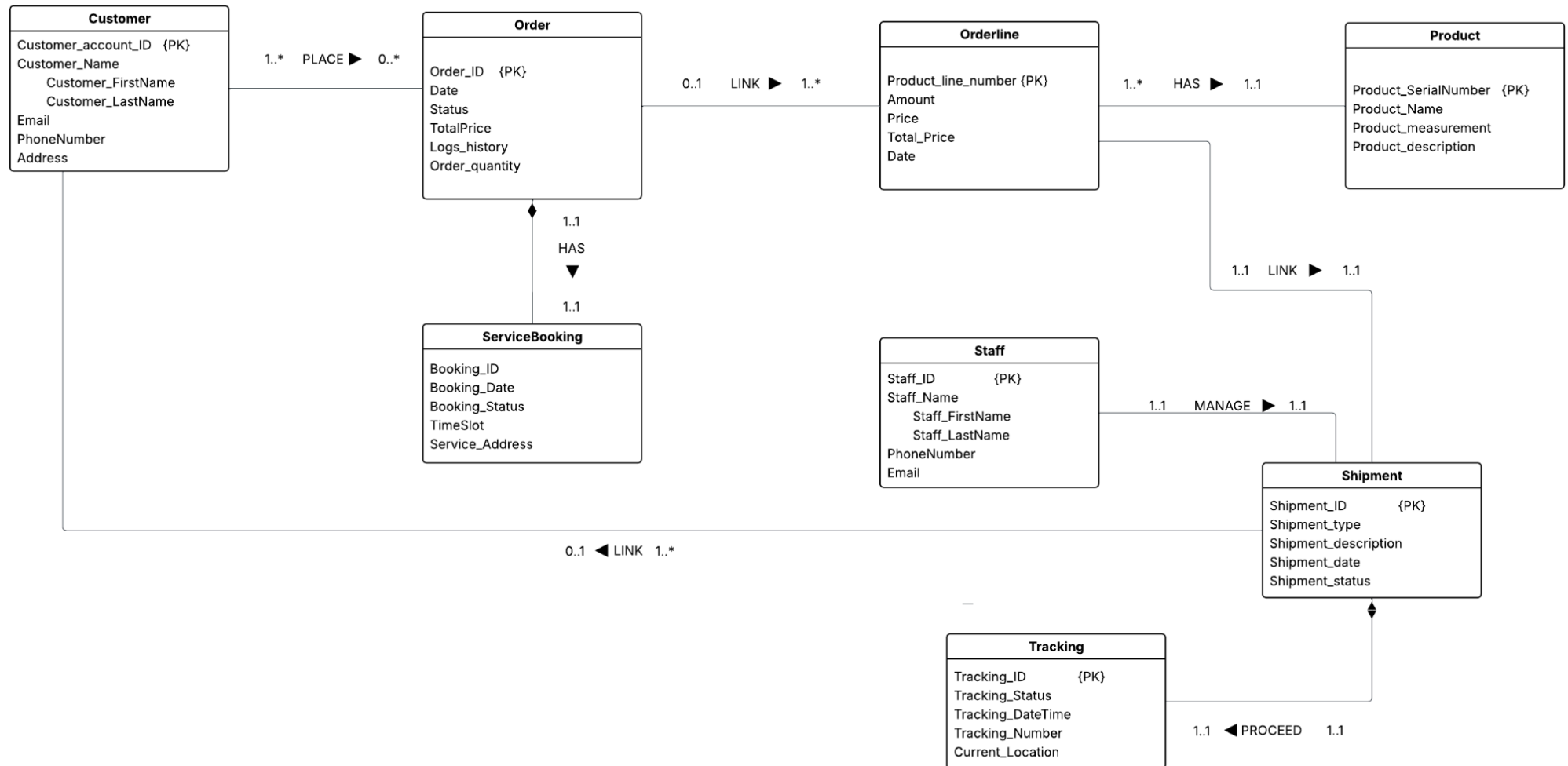
Data	Description
Shipment Id	Unique identifier for a specific delivery package.
Shipment Type	The method of delivery.
Shipment Description	Notes regarding the cargo.
Shipment Date	The date the items left the branch or stock.
Shipment Status	Real-time status of the parcel.

Tracking (Weak Entity of Shipment)

Data	Description
Tracking Id	Internal primary key for the tracking log entry.
Tracking Status	Specific milestones reached.
Tracking DateTime	The exact moment the tracking event was recorded.
Tracking Number	The external number provided by the carrier for the customer.
Current Location	The GPS coordinates or facility name where the shipment was last scanned.

IKEA UML Diagram

Ordering Management



Data Query

Example of queries required by the Staff user views:

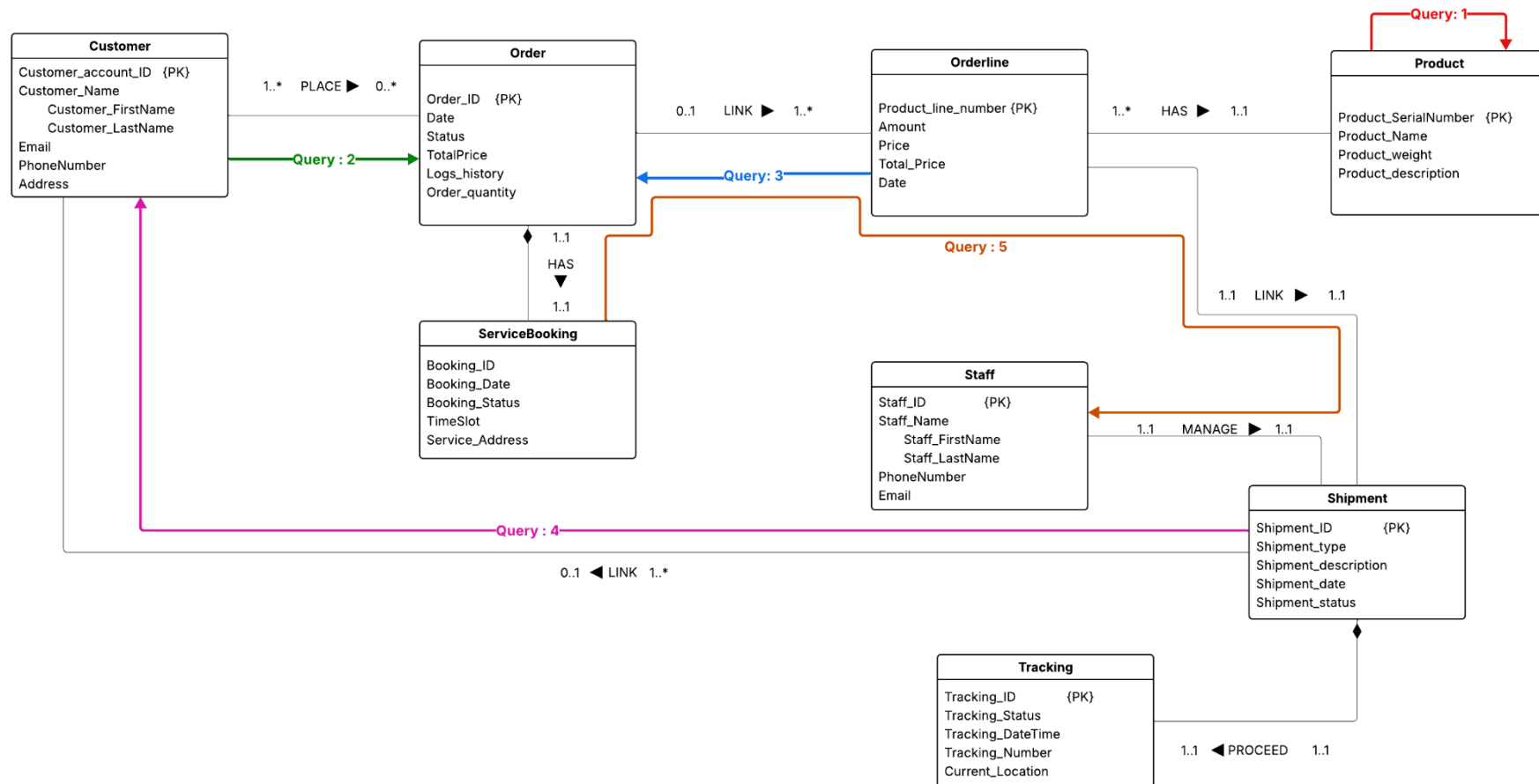
Q.1 Retrieve the name, weight, and description for a specific product using its serial number.

Q.2 Find all orders placed by a specific customer, including the date and current status.

Q.3 List every item within a specific order, including the quantity and the total price for each line item.

Q.4 Get the latest location and status updates for a customer's package.

Q.5 List all pending assembly or delivery bookings for a specific date to assign to staff.



2. Stock and Product management

Product

Data	Description
Product_SerialNumber	An unique identifier for each product.
Product_Name	A name of the product for show to customers.
Product_Description	Details of the product used for specification explaining
Product_Measurement	Product size/dimensions used for storage and delivery planning.

Stock

Data	Description
Stock_ID	An unique identifier for each inventory record.
Stock_Status	Condition and status of the stock. (Low stock, Out-of-stock, Normal)
Stock_Quantity	Current availability quantity in the store.

Warranty

Data	Description
Warranty_ID	A unique identifier for each warranty record/policy.
Warranty_Description	Details of warranty coverage and duration, including conditions and limitations.

Branch

Data	Description
Branch_ID	An unique identifier for each branch.
Branch_Name	A name of the branch for customer convenience and identification.
Branch_Address	A physical location of the branch used for the stock storage and planning for delivery
Branch_Number	Phone number of the branch used for contact, operation and communication purpose.
OperatingHours	Business hours of the branch used for services and pickup planning.

Order

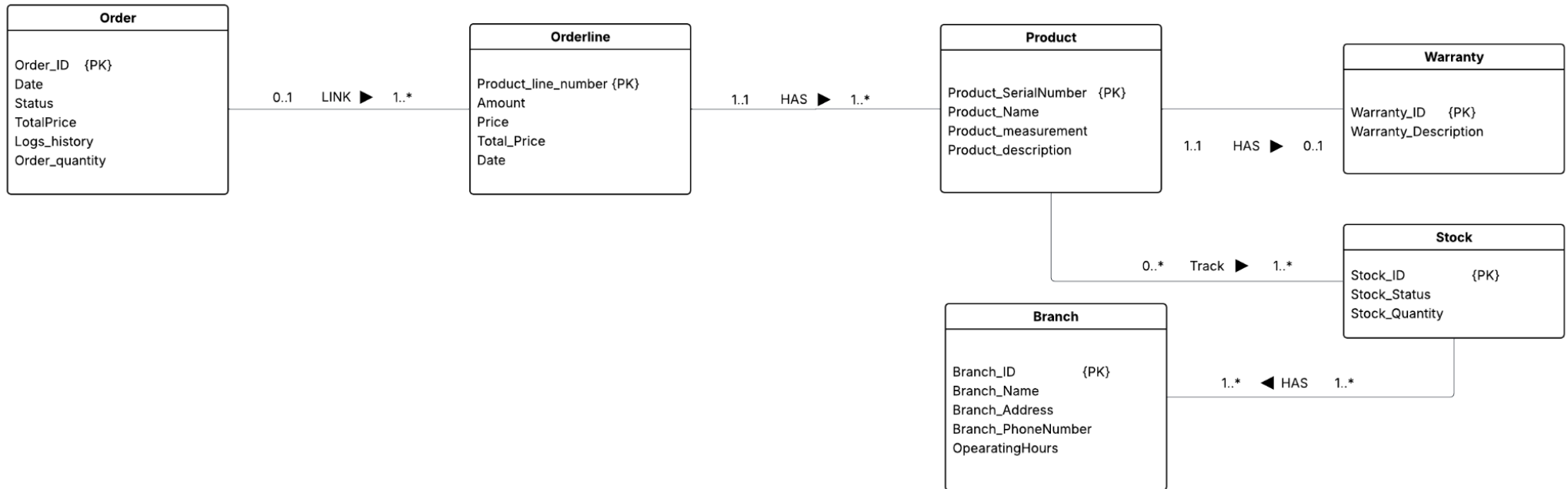
Data	Description
Order Id	A unique transaction number generated at the time of checkout.
Date	The timestamp when the order was officially placed.
Status	The current stage of the order.
Order_quantity	Total count of unique product lines included in the entire transaction.
logs_history	A chronological record of internal changes.
Total price	The final sum of all products, taxes, and service fees minus any discounts.

Orderline

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Amount	The unit price of the product at the specific time of purchase.
Total_price	The calculated subtotal for that specific line
Date	Records when that specific line item was added or modified.
Price	The price for each customer order.

IKEA UML Diagram

Stock and Product management



Data Query

Example of queries required by the Staff user views:

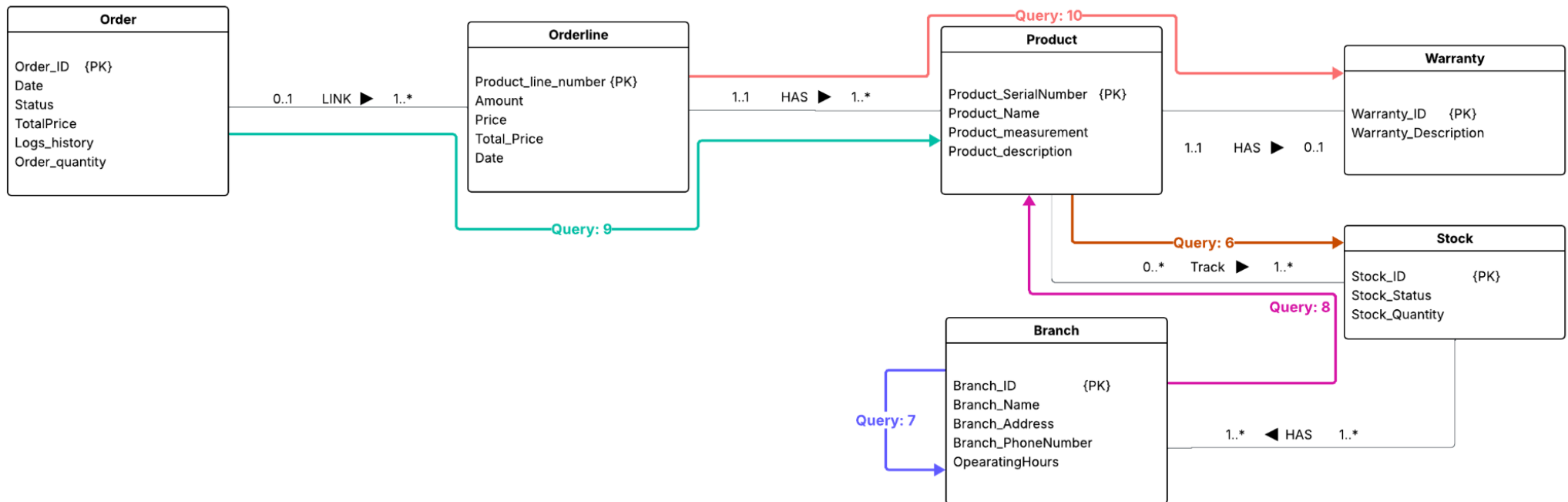
Q.6 Find the Stock_Quantity and Stock_Status for Product_SerialNumber.

Q.7 Find the total Operating hours for each branch in January 2025 by only weekdays

Q.8 Find the Product Name, Stock Id Map to their Branch

Q.9 Find the Product Name, Product Description, orderline_ID, and Order_Id for each order and orderline

Q.10 Find the Product_line_number, Product Name, and their Warranty description for each Warranty



3. Customer and Membership Care

Customer

Data	Description
Customer_account_ID	Unique identification for each Customer account.
Customer_Name (1. Customer_FirstName, 2. Customer_LastName)	Customer identity information for customer care and service support.
Address	Address information for delivery, service and account records.
Phone Number	Contact information for customer support communication.
Email	Contact information for updated information.

Membership

Data	Description
Membership_ID	Unique identification for each membership record.
Membership_tier	Tier level used to determine member benefits and discounts.
Membership_status	Status of membership while retaining membership history.

Promotion

Data	Description
Promotion_ID	Unique identification for each Promotion.
Promotion_description	Description of the promotion details, such as discount type, conditions, or special offers.
Promotion_status	Indicates the current status of the promotion (Active, Inactive, Expired).

Returns Request

Data	Description
Return_ID	Unique identification for each return request.
Return_Date	Date on which the customer is to submit the return request.
Return_Reason	The provided reason by the customer for requesting a return (such as damaged item, wrong product)
Return_Status	Current status of the return request (e.g. Pending, Approved, Rejected, Completed)
Approval_Date	The day on the return request was approved by the system or staff.
Return_Notes	Additional notes or remarks related to the return request.

Refund Request

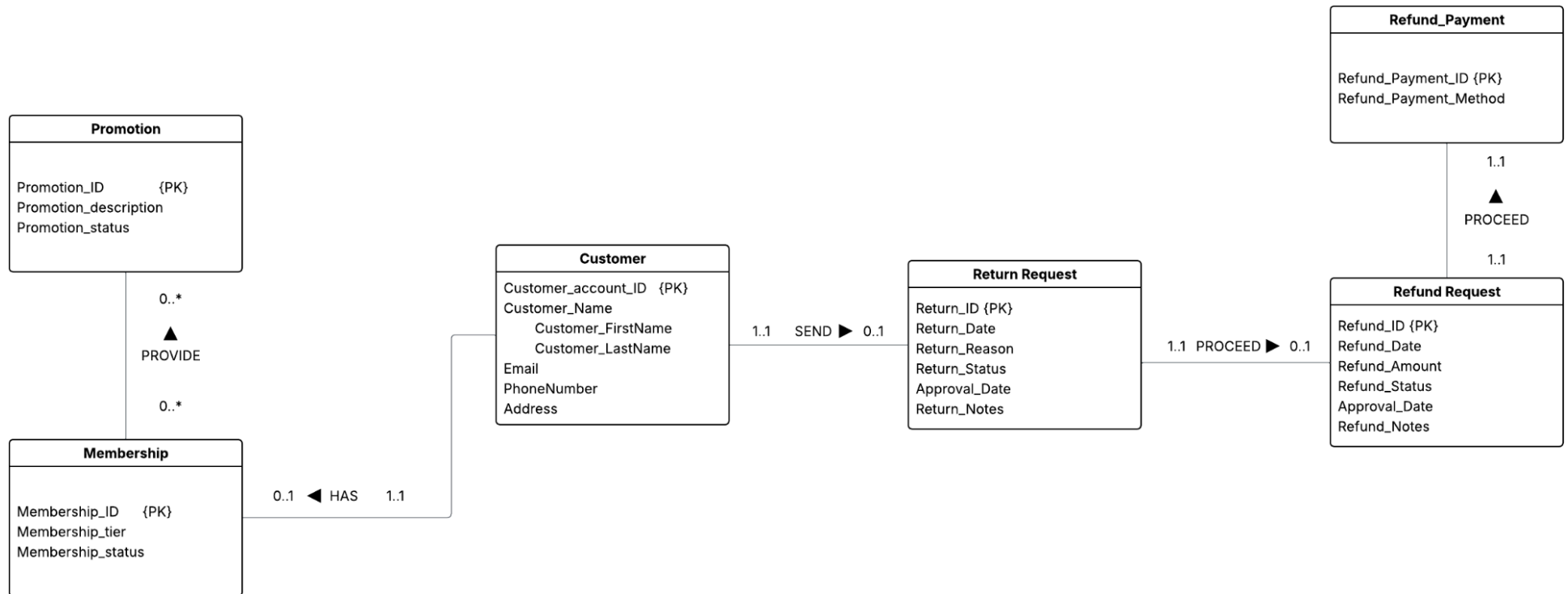
Data	Description
Refund_ID	Unique identification for each refund request.
Refund_Date	Date on which the refund request was submitted.
Refund_Amount	Amount of money to be refunded to the customer.
Refund_Status	Current status of the refund request (e.g. Pending, Approved, Rejected, Completed).
Approval_Date	The day on the refund request was approved by the system or staff.
Refund_Notes	Additional notes or remarks related to the refund request.

Refund Payment

Data	Description
Refund_Payment_ID	Unique identification for each refund payment.
Refund_Payment_Method	Method used to make the refund payment (e.g. credit card, debit card, online banking).

IKEA UML Diagram

Customer and Membership care



Data Query

Example of queries required by the Staff user views:

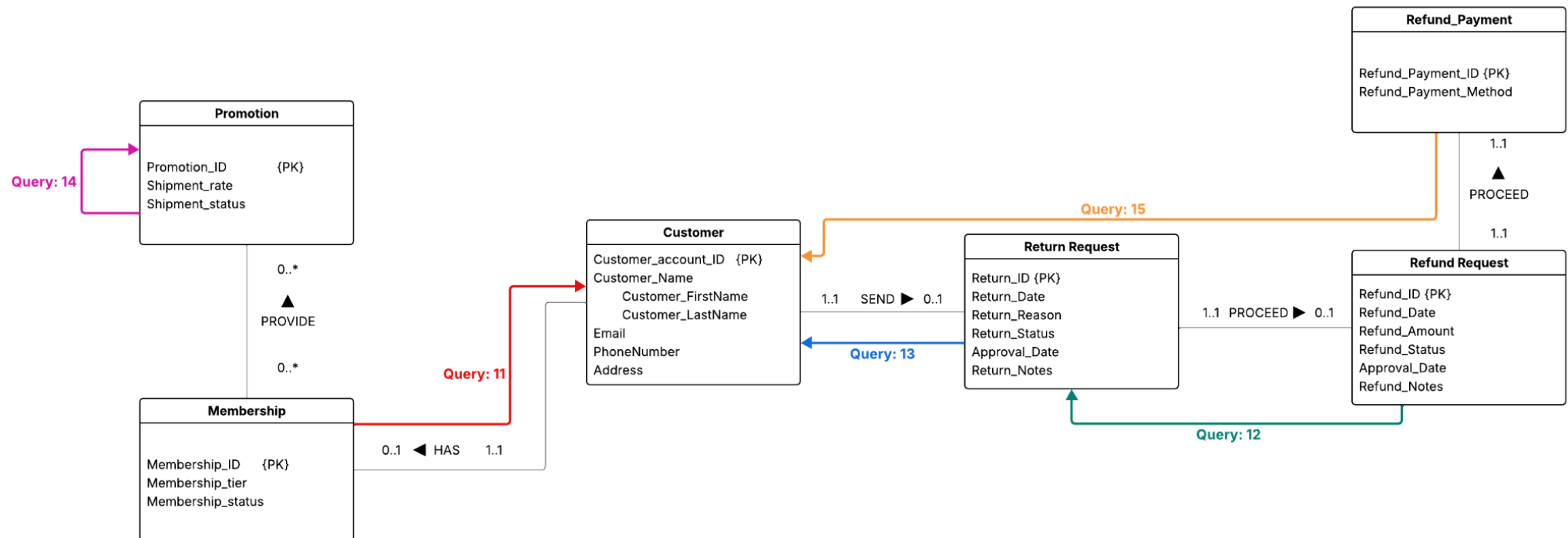
Q.11 Find the Membership_tier and Membership_status for Customer_account_ID.

Q.12 Find the Return_Status and Refund_Amount for Return_ID.

Q.13 Find the Return_Reason and Return_Notes for a help ticket submitted by a customer.

Q.14 Find all Promotions currently active for a specific Shipment_status.

Q.15 Find Refund_Amount, Refund_Payment_ID, Refund_ID, Refund_Date, Return_Reason for each customer Alphabetically.



4. Payment Management

Customer

Data	Description
Customer_account_ID	Unique identification for each Customer account.
Customer_Name (1. Customer_FirstName, 2. Customer_LastName)	Customer identity information for customer care and service support.
Address	Address information for delivery, service and account records.
Phone Number	Contact information for customer support communication.
Email	Contact information for updated information.

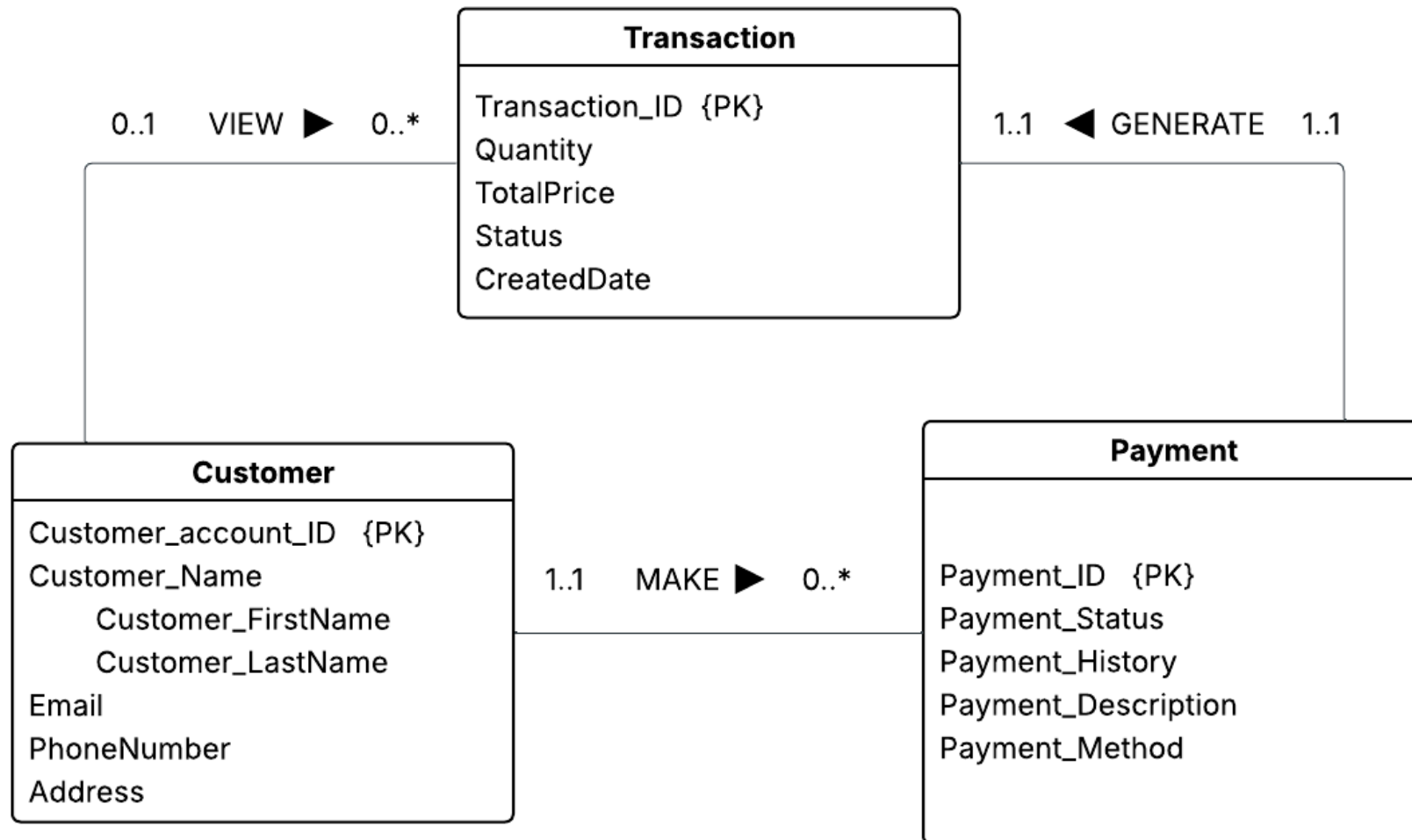
Payment

Data	Description
Payment_ID	A unique identifier for each payment record.
Payment_Status	Current payment status (Pending, Paid, Failed, Refunded).
Payment_History	A log of payment events and status changes with timestamps for tracking.
Payment_Description	Payment details such as reference info, related order, or remarks.
Payment_Method	Method used to pay

Transaction

Data	Description
Transaction_ID	A unique identifier for each payment transaction.
Quantity	Number of items included in the transaction
TotalPrice	Total amount charged in this transaction.
Status	Transaction status
CreatedDate	Date and time when the transaction was created/processed.

IKEA UML Diagram Payment Management



Data Query

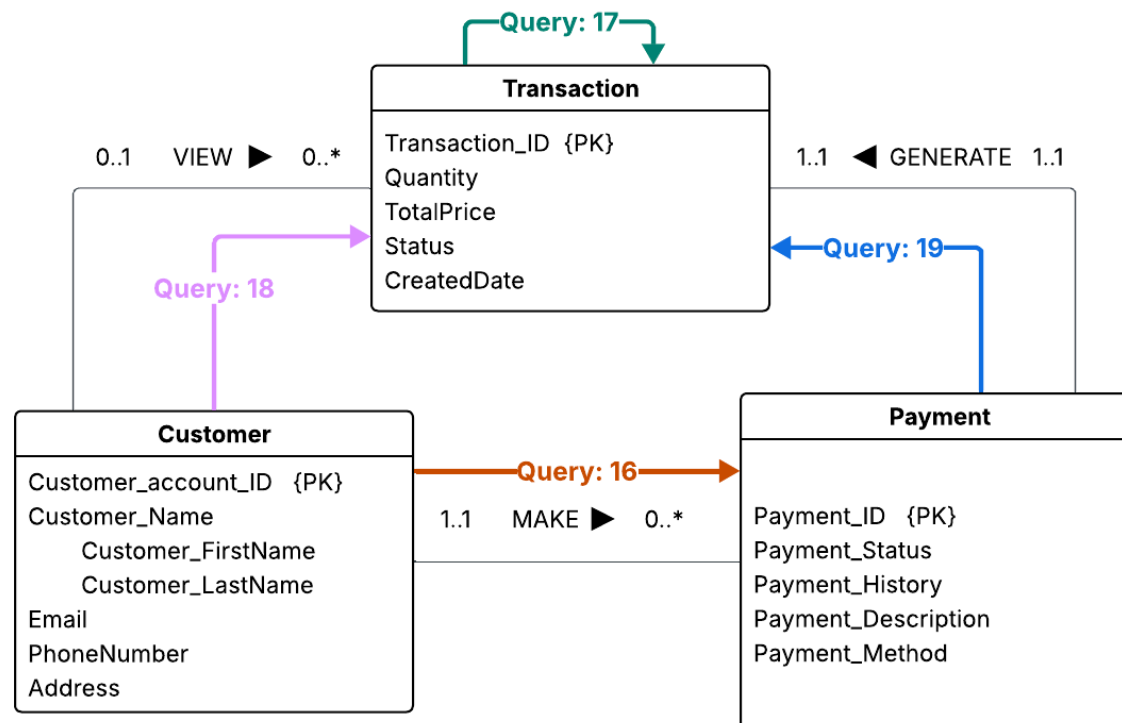
Example of queries required by the Staff user views:

Q.16 Find the payment_method and payment_description for each customer

Q.17 Find the TotalPrice and CreatedDate for Transaction_ID.

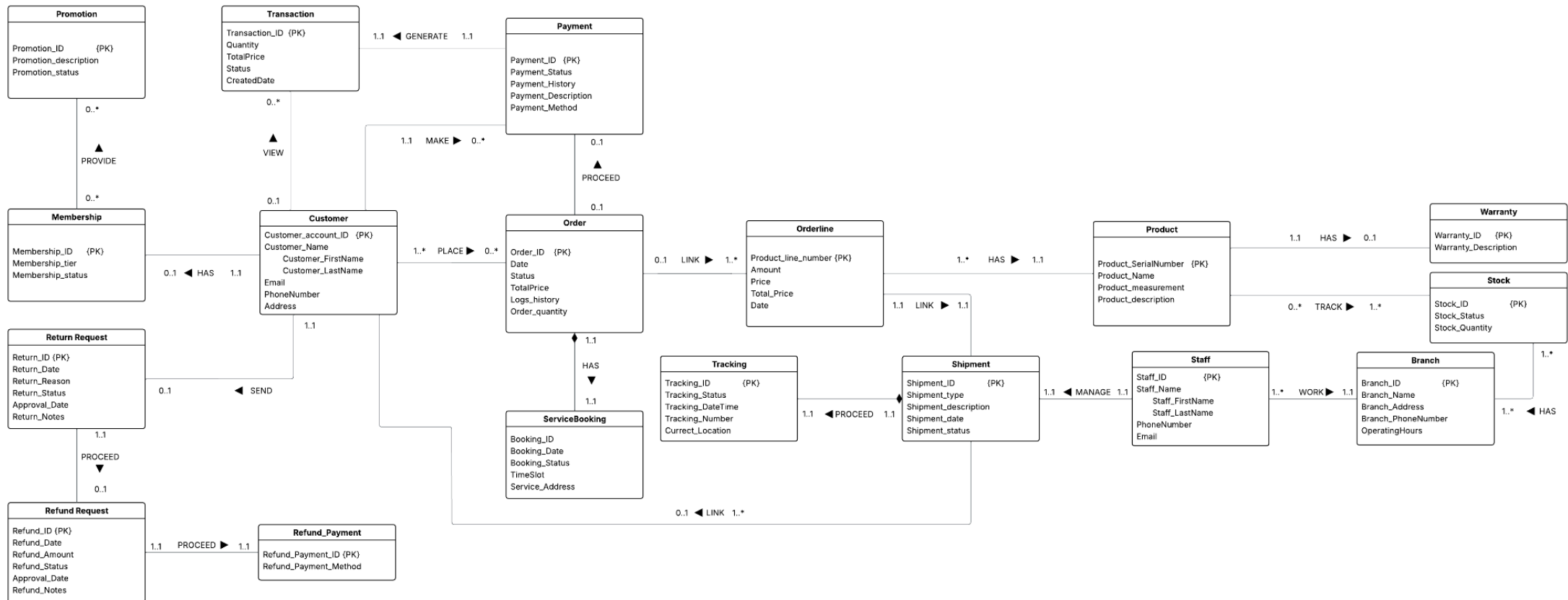
Q.18 Find all Transactions (Quantity, Status) viewed by Customer_account_ID.

Q.19 Find the History of a specific payment to verify if the Status is "Completed".



IKEA : Customer and Order Management System

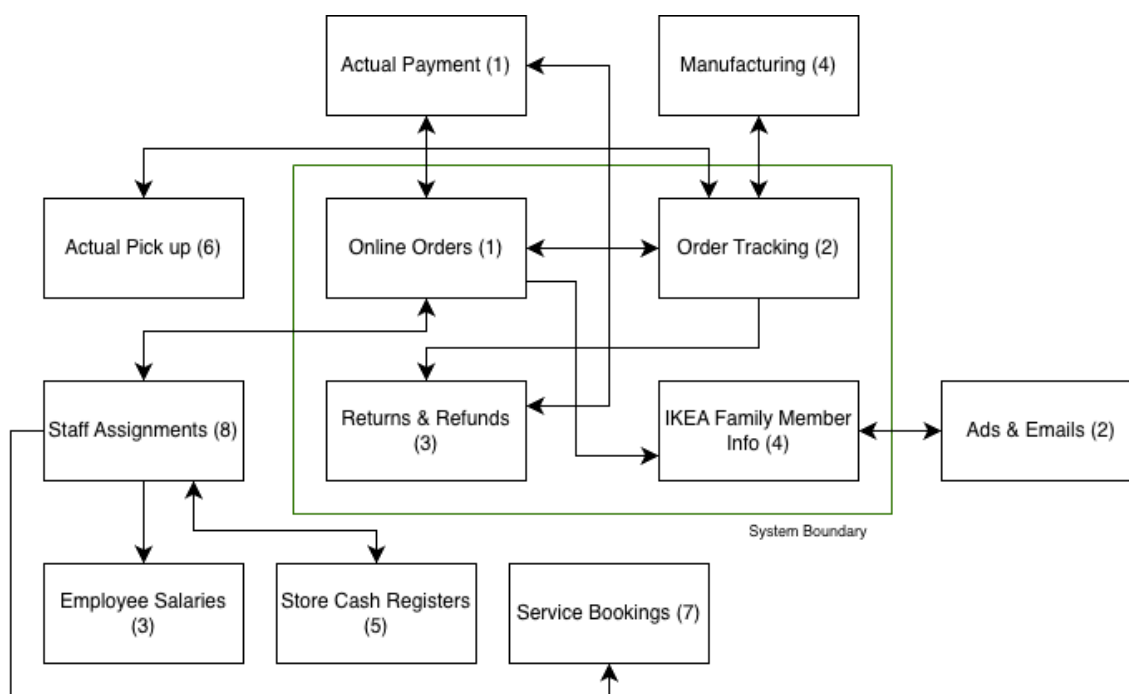
Overall UML Diagram



What will be changed from Project Phase 1

1. Scope and Boundary

Based on project 1, our group has changed some part of the scope and boundary which is moving “Staff Assignments”, “Service Bookings” out of our scope. This change was made to ensure that our database design focused on “Customer and Inventory Management” scope, which are the primary objectives of this project.



As shown in Figure 1, the system boundary of IKEA Online Store Database system is covered by four scope modules. These scope include **Online Order (1)**, which manages the core function for recording customer purchases and order details, and this module is closely integrated with **Order Tracking (2)**, which monitors order status and product availability. In addition, the system records **Returns & Refunds (3)** to track return requests and refund status, and stores **IKEA Family Member Info (4)** to support member discounts and purchase history.

Overall, these four modules work together as an integrated database system that supports online order processing, order tracking, customer membership (IKEA Family Membership Info) management, and returns and refunds handling, ensuring a smooth and consistent flow of data across IKEA’s online store operations.

In contrast, the boxes outside the green boundary represent out-of-scope systems that are not implemented within this database application. These external systems include **Actual Payment (1)**, which are processed by external payment providers, and the IKEA system only receives the payment result messages. **Ads & Emails (2)** is an external marketing system that may use IKEA Family Member data for communication. **Employee Salaries (3)** is about Human Resources Functions that are not included in the system. **Manufacturing (4)** is not covered in this database. **Store Cash Registers (5)** is used for in-store transactions. **Actual Pick up (6)** will operate when customers collect items at the store. **Service Bookings (7)** and **Staff Assignments (8)** are treated as external operational systems that interact with online orders but are not managed within this database.

Overall, the figure clarifies that this project focuses on centralizing IKEA Thailand’s core service data for online operations while treating in-store activities as an external system.

2. Data Requirements

Product

The data stored on each product includes product name, product serial number, product description, and product dimension so the customers can browse and purchase items correctly. The system must store inventory data such as stock quantity and stock status for checking stock availability for a “Click & Collect” order and must prevent operational error in the ordering process when the product is out of stock.

Order

The data stored on each online order includes order ID, data, status, total price, logs history, and order quantity. The database also needs to keep order tracking status logs to support the timeline and history of updates and manage the order process from the initial process to completed process.

Customer

The data stored on each customer includes customer account ID, customer name(first name, last name), email, phone number, address and status of IKEA Family membership so the customers can place orders, receive services, and track their orders.

Payment

The data stored on each payment include payment id, order id, product name, customer name, payment history, and payment status to make the accurate tracking and verification of payment transactions and order processing. Each payment must be linked to one specific order. Payment status must be recorded as “successful”, “failed”, or “refunded”. Payment information must not be changed after the payment status has been finalized. Each payment must record only one payment method.

Membership

The data stored on each membership includes membership id, customer name, and membership tier. Each membership must have a unique membership ID for identification. Each membership must have only one tier at any given time. A customer may hold only one active membership at a time. Membership information must be retained in the system after the membership has expired.

Staff

The data stored on each staff includes name, address, phone number, and email. Each staff email address must be unique in the system. Each staff information may only be updated or changed by the staff member who owns the record.

3. Transaction Requirement

Data Approach

- Enter the details of a new online order that is created from a customer cart.
- Enter the details of a payment result that is returned from the payment of the specific order.
- Enter the details of a service booking to an order.

Data Update / Deletion

- Update / delete the details of an online order.
- Update the payment status of an order based on the payment status and store payment history.
- Update the stock quantity of a product after an order is being placed, confirmed, and update the stock when the item is being returned and refunded.
- Update service booking details for an order.
- Update the return and refund status and delete the duplicated or invalid return request.

4. Data Queries

Query 1-5 : Ordering Management

- Q.1 Retrieve the name, weight, and description for a specific product using its serial number.
- Q.2 Find all orders placed by a specific customer, including the date and current status.
- Q.3 List every item within a specific order, including the quantity and the total price for each line item.
- Q.4 Get the latest location and status updates for a customer's package.
- Q.5 List all pending assembly or delivery bookings for a specific date to assign to staff.

Query 6-10 : Stock and Product Management

- Q.6 Find the Stock_Quantity and Stock_Status for Product_SerialNumber.
- Q.7 Find the total Operating hours for each branch in January 2025 by only weekdays.
- Q.8 Find the Product Name, Stock Id Map to their Branch.
- Q.9 Find the Product Name, Product Description, orderline_ID, and Order_Id for each order and orderline.
- Q.10 Find the Product_line_number, Product Name, and their Warranty description for each Warranty.

Query 11-15 : Customer and Membership Care (IKEA Family)

- Q.11 Find the Membership_tier and Membership_status for Customer_account_ID.
- Q.12 Find the Return_Status and Refund_Amount for Return_ID.
- Q.13 Find the Return_Reason and Return_Notes for a help ticket submitted by a customer.
- Q.14 Find all Promotions currently active for a specific Shipment_status.
- Q.15 Find Refund_Amount, Refund_Payment_ID, Refund_ID, Refund_Date, Return_Reason for each customer Alphabetically.

Query 16-19 : Payment Management

- Q.16 Find the payment_method and payment_description for each customer.
- Q.17 Find the TotalPrice and CreatedDate for Transaction_ID.
- Q.18 Find all Transactions (Quantity, Status) viewed by Customer_account_ID.
- Q.19 Find the History of a specific payment to verify if the Status is "Completed".

What will be changed from Project phase 1

In summary, our group has fixed some part from the Project 1 phase 1 that will be added into Project 1 phase 2 include “Scope and Boundary”, “Data Requirement”, “Transaction Requirement”, “Data Queries”, “Entity and Attribute Implementation”.

1. Scope and Boundary

- Move “Staff Assignments” and “Service Bookings” out of scope of our system and fix the UML diagram description.

2. Data Requirement

- Fix a description of “Product”, “Order”, “Customer”, “Payment”, “Membership”, “Staff” on each Attribute.

3. Transaction Requirement

- Fix the description of “**Data approach**” and “**Data Update / Deletion**” based on UML Schema.

4. Data Queries

- Divide the Data Queries into 4 parts following:
 - “Query 1-5 : Stock and Product Management”
 - “Query 6-10 : Order Management”
 - “Query 11-15 : Customer and Membership Care”
 - “Query 16-19 : Payment Management”

5. Entity and Attribute Implementation

- Add the Entities from the following: “Branch”, “Orderline”, “Return Request”, “Refund Request”, “Refund Payment”, “ServiceBooking”(Weak Entity of Order), “Shipment”, “Stock”, “Tracking”(Weak Entity of Shipment), “Transaction”, and “Warranty”.

Reference

Formal UML Logic (Standard Rules) : <https://www.omg.org/spec/UML/>

Industry Reference: IKEA Thailand – Customer Service and Returns Policy:
<https://www.ikea.com/th/en/customer-service/>.

ITCS442_Relational Database Design lect 1-3 Faculty of Information and Communication Technology Mahidol University : Instructor: Asst.Prof. Dr. Charnyote Pluempitiwiriyawej.

AI level 2 from Google Gemini : <https://gemini.google.com>

Operations management of IKEA [SlideShare.com](https://www.slideshare.net/slideshow/operations-management-of-ikea/187051571) :
<https://www.slideshare.net/slideshow/operations-management-of-ikea/187051571>

User registration form <https://family.ikea.co.th/registration>

Employee application:
<https://jobs.smartrecruiters.com/IkanoRetail/744000102921265-business-analyst-finance-business-partner-commercial-finance-ikea-sukhumvit-and-ikea-bangna?trid=c39a8cee-af0e-4db1-9034-acbba85a4a4d>

Delivery service report and Tracking online orders report:
<https://www.ikea.com/mx/en/customer-service/track-manage-order/>

IKEA: A Case Study Through the Lens of Algorithms and Data Structures:
<https://medium.com/@harshada.deshingkar22/ikea-a-case-study-through-the-lens-of-algorithms-and-data-structures-af8de39e47b7>

Case Study Ikea: <https://www.scribd.com/document/468047669/Case-Study-Ikea>

System and boundary diagram from draw.io:
https://drive.google.com/file/d/1UqdvIA_ltw7rvLBpeILYLLjwyldW-z_3/view?usp=sharing

UML Diagram and Drafting by Lucid.app:
https://lucid.app/lucidchart/4fb7d03f-d529-43b3-900d-0257bfbe9b52/edit?viewport_loc=-1180%2C-1825%2C10702%2C6397%2CisStbOByeoYb&invitationId=inv_f68534b1-f4bf-4396-adc4-c0e7b2ca4d65